Comparative Evaluation of Custom CNN, and ResNet50V2 Transfer Learning, for Alzheimer's Disease Classification

Abstract

Alzheimer's disease is a progressive neurodegenerative disorder characterised by cognitive decline and structural brain atrophy, which can be visualised using magnetic resonance imaging (MRI). Accurate classification of disease severity remains challenging, particularly in early stages. This study developed and compared two convolutional neural networks (CNNs), for the automated classification of 5,076 brain MRI scans into three categories: "No Impairment", "Mild Impairment", and "Moderate Impairment". Images were pre-processed through resizing, normalisation, and contrast enhancement (CLAHE), with stratified data splitting, data augmentation and class weightings applied to address imbalance between classes. A custom CNN was evaluated against a transfer learning model, based on ResNet50V2 pre-trained on ImageNet. Performance was assessed using precision, recall, F1-score, confusion matrices, receiver operating characteristic (ROC) and accuracy/loss curves. The transfer learning model demonstrated a stronger performance in classifying disease states, achieving an overall F1-score of 82% and a macro-average AUC of 0.933, compared to 61% and 0.846 for the custom CNN. It also showed improved sensitivity for minority impairment classes, and stable convergence during model training. Overall, these findings demonstrate that transfer learning enhances robustness and diagnostic reliability in MRI-based Alzheimer's classification, reinforcing its potential in supporting clinical decision-making in a healthcare setting.

Introduction

Dementia is an umbrella term that describes a collection of neurodegenerative disorders characterised by progressive cognitive decline, causing memory impairment (Jean and Dotson, 2024). Alzheimer's disease (AD) is the most prevalent form of dementia, accounting for approximately two-thirds of global cases in individuals aged 65 and older (Kumar *et al.,* 2024). AD is distinguished by the accumulation of beta-amyloid plaques outside neurons, followed by formation of tau tangles within neurons. This impairs synaptic function and neuronal integrity, typically affecting the hippocampus and medial temporal lobe structures (Breijyeh and Karaman, 2020). Progressive neuronal loss in these brain regions can lead to cortical and hippocampal atrophy that can be detectable through structural imaging techniques (Sabuncu *et al.,* 2011).

Magnetic Resonance Imaging (MRI) is widely used to visualise structural brain abnormalities and characteristic atrophy patterns across stages of AD, supporting clinical diagnosis (Živanović et al., 2023). However, diagnosis remains challenging as early-stage AD shows subtle and non-specific structural changes that can overlap with normal aging and other dementias, limiting its reliability as a stand-alone diagnostic tool (Aramadaka *et al.,* 2023). Such limitations highlight the need for more objective and data-driven diagnostic approaches.

Artificial intelligence and machine learning (AI/ML) provide a promising alternative. By learning patterns from large imaging datasets, ML models can assist clinicians in identifying and classifying disease stages of Alzheimer's with greater consistency (Iglesias *et al.,* 2021; Kufel *et al.,* 2023). Artificial neural networks (ANNs) are a class of ML models inspired by the

biological neural system, consisting of interconnected layers of computational units that learn by adjusting weighted connections to minimise prediction error (Kufel *et al.,* 2023).

Convolutional Neural Networks (CNNs) are a specialised subclass of ANNs, designed specifically for image analysis (Yamashita *et al.,* 2018). Through convolutional layers that apply learnable filters across input images, CNNs extract spatial hierarchical features including textures, edges and structural patterns (Ali *et al.,* 2025). Its ability to learn important structural patterns from images makes it a perfect candidate for classifying MRI disease stages of AD.

This study develops and evaluates two CNN based models trained on structural MRI data, to classify scans into three disease states: “No Impairment”, “Mild Impairment” and “Moderate Impairment”. This report will cover the methodology, evaluate and compare model performance, and discuss clinical and ethical implications of integrating AI-driven tools in a healthcare setting.

Materials and Methods

Data Visualisation

All analyses and model development were conducted in Google Colab, using Python 3.12. A simple schematic of the overall methodological workflow can be seen in Figure 1.

The dataset comprised of 5,076 MRI images (jpg), categorised into three disease states: “No Impairment” (n=2,560), “Mild Impairment” (n=1,792) and “Moderate Impairment” (n=724). Class distribution was visualised using a bar chart (Fig.2) revealing a clear imbalance between classes, with “No Impairment” representing 50.4% of the dataset, “Mild Impairment” 35.3%, and “Moderate Impairment” 14.3%. Class imbalance can bias the model training toward the majority class, potentially reducing sensitivity for minority disease stages.

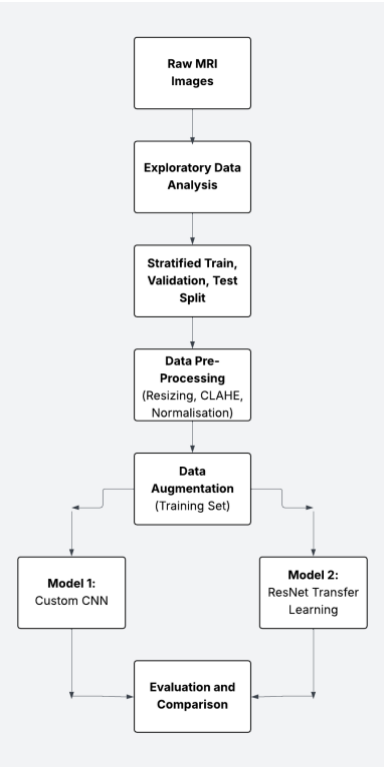


Fig 1. Methodological Workflow for MRI Image Classification:
Schematic overview of the projects' pipeline (created through LucidChart). The workflow consists of importing and visualising raw data, preprocessing, augmentation and model development. Two architectures were implemented: a custom CNN and a ResNet based transfer learning model. The performance of these models was then evaluated and compared through classification metrics and subsequent plots.

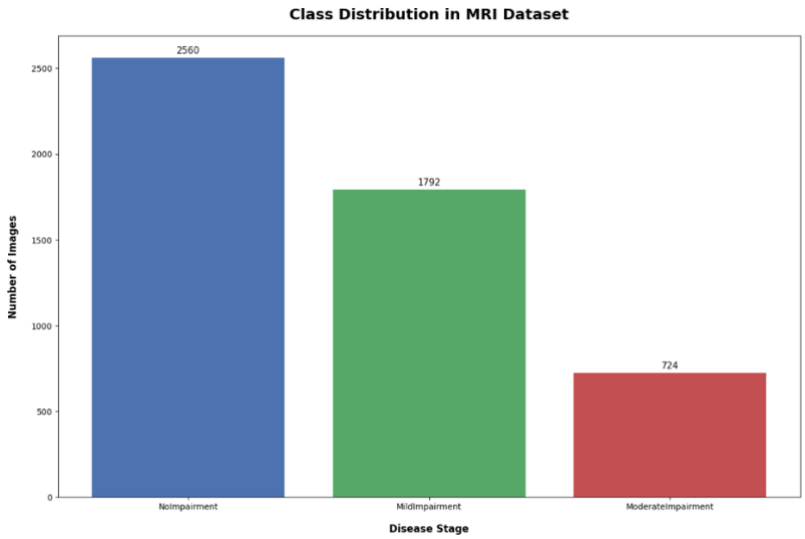


Figure 2. Class Distribution of the MRI dataset.
A bar chart displaying the number of images per disease state. "No Impairment" (n=2,560), "Mild Impairment" (n=1,792), and "Moderate Impairment" (n=724). This bar chart demonstrates class imbalance amongst the three disease states, with most of the MRI image samples belonging to the "No Impairment" class.

Data Splitting and Preprocessing

The dataset was split using a stratified approach across all disease states. Initially, 20% of the data was reserved as an independent, unseen test set for final evaluation. The remaining 80% was further divided, allocating 64% to the training data and 16% for validation. The training set was used to teach the model to learn patterns from the data, while the validation set helps tune hyperparameters and monitor overfitting (Yamashita *et al.,* 2018). All preprocessing parameters were fitted strictly to the training data to prevent data leakage, mitigating risk of artificially inflating model performance (Rosenblatt *et al.,* 2023).

During preprocessing, images were resized to 224x224 pixels, rescaled to a [0,1] range to stabilise gradient descent (optimisation algorithm that iteratively minimises model error (Zhang *et al.,* 2019)). To enhance the visibility of subtle features present in the MRI images, Contrast Limited Adaptive Histogram Equalization (CLAHE, clip limit=2.0) was applied.

Following preprocessing, two distinct models were developed: a custom CNN (Model 1) and a transfer learning model (Model 2)

Model 1: Custom CNN

The first model was developed from scratch using a sequential CNN architecture. To capture initial low-level spatial features, the network begins with standard Conv2D layers (64 filters) followed by 4 deep convolutional blocks (128, 256, 512 and 728 filters) to enable hierarchical feature extraction of MRI images. To maximise parameter efficiency, these deep blocks utilise “SeperableConv2D layers”. Each block contains ReLU activation function (introduces non-linearity during training), followed by batch normalisation and max-pooling for spatial down sampling (Hartmann *et al.,* 2021).

The convolutional feature maps were flattened using global average pooling and passed through fully connected dense layers prior to classification of disease. Progressive dropout regularisation (rate=0.2-0.5) was incorporated to reduce overfitting by randomly deactivating neurons during training (Shehab and Alhaddad, 2025). The final layer consisted of three neurons with softmax activation to produce class probabilities corresponding to their respective disease state.

To improve generalisation, data augmentation was carried out to the training set only. Augmentation included horizontal flipping and small random rotations, zoom, contrast, adjustment and translation (5% range). This increases training variability and reduces risk of overfitting.

Given the observed class imbalance (Fig.2), balanced class weights were implemented during training using scikit-learns “compute_class_weight” function. This assigns higher weights to under-represented classes and lower weights to majority class based on their frequency in the training data. The resulting weights were 0:0.66 (“No Impairment”), 1:0.94 (“Mild Impairment”), and 2:2.33 (“Moderate Impairment”). These weightings encourage balanced learning across classes by penalising misclassification of minority classes more heavily.

The model trained using a learning rate of 0.001 and optimised with categorical cross-entropy loss. Training was performed using a batch size of 32 for up to 50 epochs.

Model 2: Transfer Learning with ResNet50V2

The second model utilised transfer learning using ResNet50V2, a 50-layer deep CNN pre-trained on the ImageNet dataset containing millions of labelled images across thousands of categories (Hossain *et al.,* 2022). Transfer learning repurposes CNNs already pre-trained on large scale datasets (ImageNet) for medical image classifications by reusing learned visual features and retraining selected layers, improving model performance and reducing training time (Kim *et al.,* 2022).

The MRI images were lightly augmented to the training set (5% random zoom and contrast), and rescaled to [-1,1]. The model also utilised pre-trained weights from ImageNet.

The convolutional base was followed by a custom classification head comprising global average pooling, two fully connected dense layers (256 and 128 units), with ReLU activation and dropout (rate=0.3). A final softmax layer was added for class classification.

Training was conducted in two stages. During the initial warmup phase, the pre-trained convolutional base was frozen and only the classification head was trained for 10 epochs (learning rate=1e-4). In the fine-tuning phase, the top 20 layers of ResNet50V2 base were unfrozen to adapt pre-learned features to MRI data. Fine-tuning was trained for 20 epochs, with a reduced learning rate of 1e-5 to prevent destabilisation of previously learned weights.

Results

Overall Performance and Classification Metrics

Model 2 significantly outperformed Model 1 across all evaluation metrics (Fig.3). Model 1 achieved 61% test accuracy with a weighted F1-score of 0.61, whereas Model 2 reached 82% accuracy with a weighted F1-score of 0.82. Macro-average F1-score increased from 0.58 in Model 1 to 0.81 in Model 2, indicating Model 2 improved class-balanced performance even in the presence of an imbalanced dataset.

Model 1 struggled with minority classes (Fig.3-4), with F1-scores of 0.49 (“Mild”), 0.54 (“Moderate”) and 0.71 (“No Impairment”). In contrast, Model 2 showcased balance across all categories, yielding F-1 scores of 0.85 (“No Impairment”), 0.79 (“Mild”) and 0.78 (“Moderate”).

a.

Model 1: Classification Report

	precision	recall	f1-score	support
NoImpairment	0.76	0.68	0.71	512
MildImpairment	0.48	0.51	0.49	358
ModerateImpairment	0.49	0.59	0.54	144
accuracy			0.61	1014
macro avg	0.58	0.59	0.58	1014
weighted avg	0.62	0.61	0.61	1014

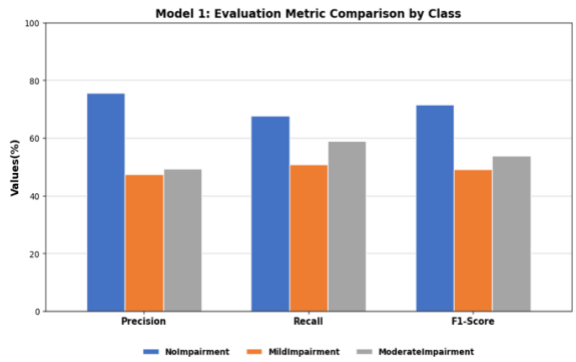
b.

Model 2: ResNet50V2 Classification Report

	precision	recall	f1-score	support
NoImpairment	0.89	0.82	0.85	512
MildImpairment	0.78	0.80	0.79	358
ModerateImpairment	0.72	0.85	0.78	144
accuracy			0.82	1014
macro avg	0.79	0.83	0.81	1014
weighted avg	0.82	0.82	0.82	1014

Fig 3: Classification reports of Model 1 and 2
Precision, recall, F1-score, and overall accuracy are shown per class. Precision reflects the proportion of predicted positive cases that are correctly identified, while recall represents the proportion of actual positive cases correctly detected. Accuracy represents overall correct predictions across all classes. The macro averages represent the unweighted mean performance across classes, whereas weighted average accounts for class frequency. Model 2 demonstrates an improved and more balanced classification performance across all Alzheimer’s disease stages.

a.



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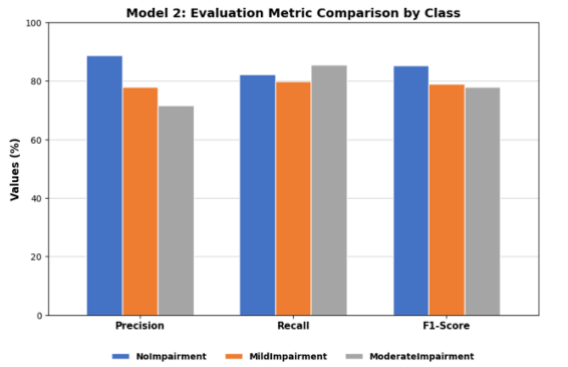
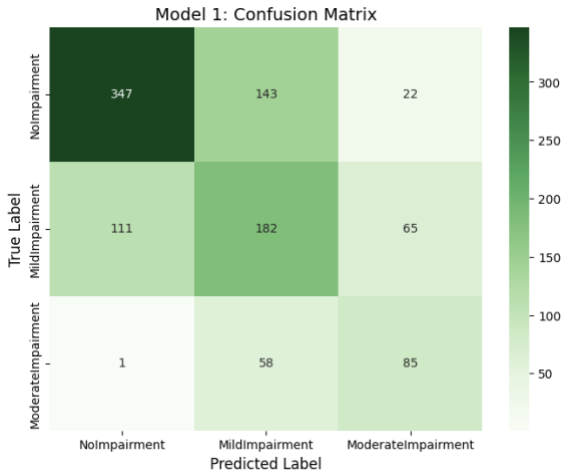


Fig 4: Comparative Evaluation Metrics by Disease State
These bar charts illustrate the precision, recall and F1-scores for the “No Impairment” (blue), “Mild Impairment” (orange) and “Moderate Impairment” (grey). Model 1 (a) highlights poor predictive accuracy for the minority impairment classes, whereas Model 2 (b) demonstrates a more robust and consistent classification across all three disease states

Confusion Matrix Analysis

The confusion matrices (Fig.5) provide insight into classification errors between the models. Model 1 struggled considerably with class overlap, frequently misclassifying true “Mild Impairment” cases as either “No Impairment” (111 instances) or “Moderate Impairment” (65 instances). Model 2 reduced these errors, correctly identifying 286 “Mild Impairment” cases and 123 “Moderate Impairment” cases, resulting in a stronger concentration of true positives along the diagonal.

a.



b.

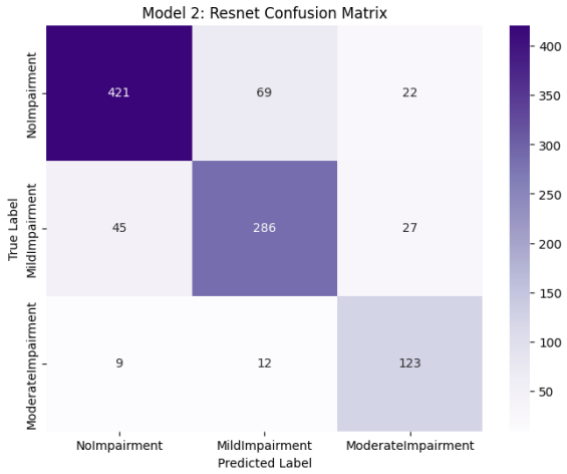


Fig 5: Confusion matrices, evaluating predictive performance across disease states
Model 1 (a) illustrates a high rate of misclassifying disease states of MRI images and shows that the model may be struggling to tell the difference apart across the three disease states. Model 2 (b) displays a superior diagnostic accuracy to Model 1, with a high concentration of true positive predictions along the diagonal.

Training

Figures 6-7 demonstrate unstable convergence in Model 1, showcased by pronounced fluctuations in validation accuracy and spikes in validation loss (especially around epoch 10). Model 2 however, displayed a smoother convergence across both figures. At epoch 10, the initiation of fine-tuning caused a temporary disturbance in both validation accuracy and loss. Despite this, the model expectedly recovered due to the application of transfer learning, with validation accuracy steadily climbing to approximately 80% and loss consistently decreasing. However, Model 2 shows mild but controlled overfitting.

a. b.

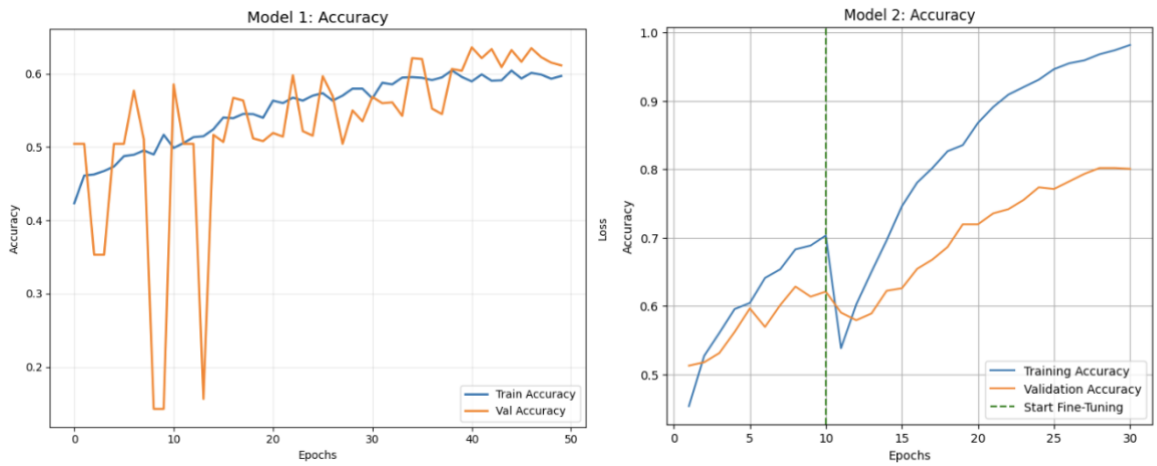


Fig 6: Comparative training and validation accuracy trajectories
Model 1 (a) accuracy progressed across 50 epochs, with significant volatility and unpredictable validation performance. This means the training was unstable, leading to a poor convergence between train and validation accuracy. Model 2 (b) accuracy progression across 30 epochs and demonstrated stable convergence following initiation of fine-tuning at epoch 10. However, as training accuracy reaches 1.0, validation accuracy is roughly 0.8, indicating presence of mild overfitting

a. b.

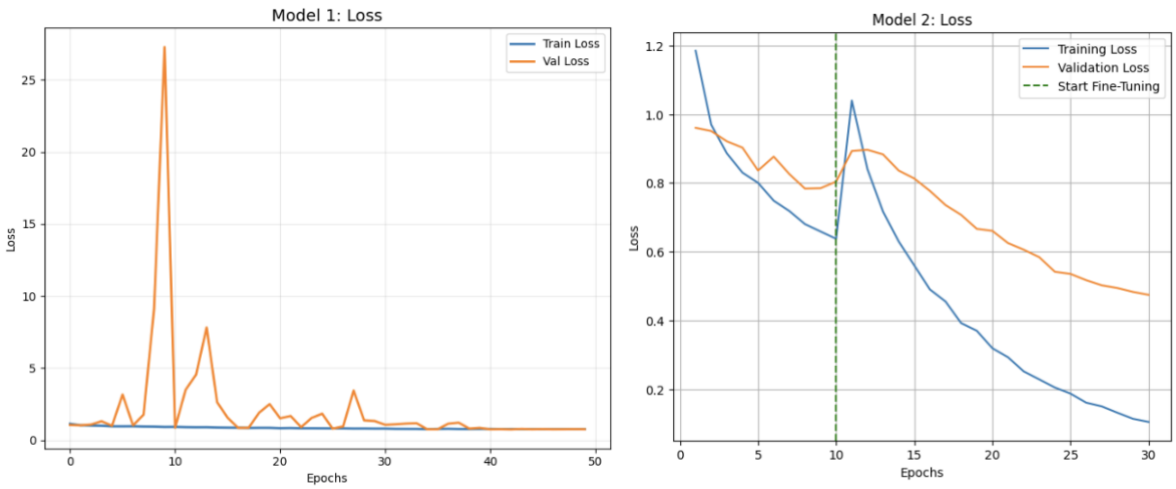


Fig 7: Comparative training and validation loss trajectories
Model 1 (a) loss progression across 50 epochs, characterised by pronounced spikes in validation loss, indicating training instability. Model 2 (b) loss progression across 30 epochs, showing a consistent downward trend and stable convergence. The temporary disturbance at epoch 10 corresponds to the fine-tuning phase, after which the model successfully recovers and continues to minimise error. However, as training loss reaches near 0, validation accuracy reaches around 0.45, indicating presence of mild overfitting.

Receiver Operating Characteristic (ROC) analysis

Figure 8 highlights an ROC curve illustrating sensitivity (proportion of actual positive cases correctly identified) versus false positive rate (proportion of negative cases incorrectly identified as positive) (Nahm *et al.,* 2022). Model 2 achieved a Macro-Average AUC of 0.933, indicating a stronger discriminative ability in classifying disease state of AD, compared to Model 1's AUC of 0.846.

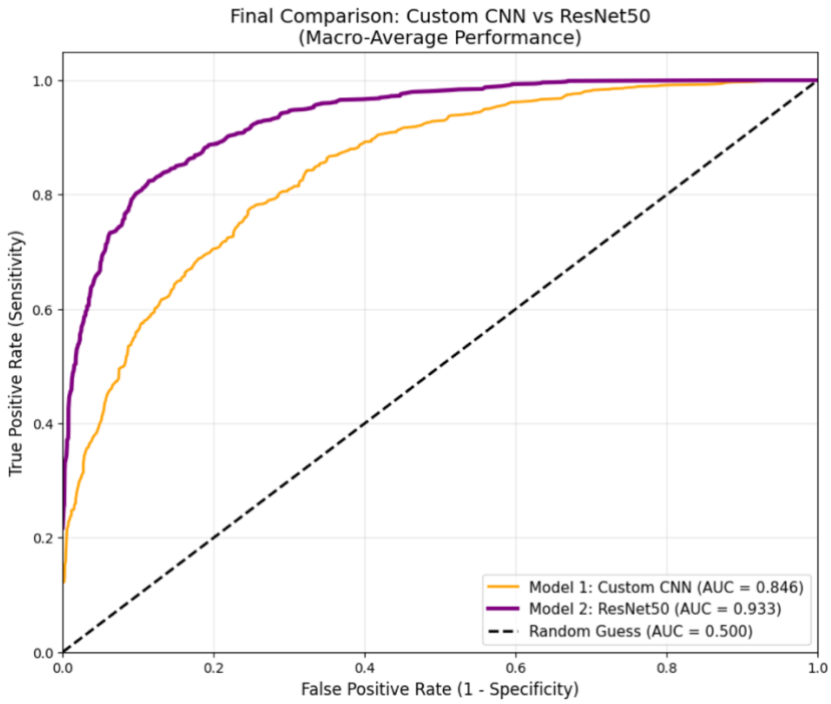


Figure 8: Comparative Receiver Operating Characteristic (ROC) curves.
ROC curve comparing how well both models can distinguish between the three disease states, with true positive rate (sensitivity) on the Y axis and false positive rate (1-specificity) on the X axis. The Area Under the Curve (AUC) represents overall diagnostic accuracy of the models, where a perfect score is 1.0, and a random guess is 0.5. Model 2 (purple) got an AUC of 0.933, compared to Model 1 (yellow) which got an AUC of 0.846.

Ethical Implications

AI/ML has the potential to enhance diagnostic accuracy, particularly through advanced medical imaging analysis (Fahim *et al.,* 2025). By identifying subtle patterns that may not be discernible through conventional approaches, ML models can support earlier detection of disease and improve diagnostic precision that may be difficult to discern through visual inspection alone (Hunik *et al.,* 2026).

Furthermore, predictive models can identify high-risk individuals and forecast disease progression, facilitating early interventions that can reduce mortality and late-stage treatment costs (Wang *et al.,* 2025). As the volume of medical data rises, integrating AI/ML diagnostic predictions can significantly reduce administrative burnout in the NHS, facilitating more efficient and standardised clinical decision-making (Pavuluri *et al.,* 2024).

However, the deployment of deep learning systems raises significant ethical concerns. Deep learning AI models are often called “black boxes”, due to their opaque decision-making processes (Qamar and Bawany, 2023). This lack of transparency makes predictions difficult to interpret and justify, limiting the physicians’ ability to explain model outcomes, detect errors and assign accountability (Marcus and Teuwen, 2024)

Consequently, AI misdiagnoses raise ethical concerns regarding unclear accountability, as responsibility of mistakes may fall on developers, clinicians or healthcare organisations (Corn, 2023). Such errors can lead to delayed or inappropriate treatment plans, affecting patient health (Habli *et al.,* 2020).

Moreover, algorithmic bias from unrepresented training data (particularly across race/gender) may perpetuate health inequities and lead to inaccurate risk estimation in vulnerable populations (Chustecki, 2024).

Ethical use of AI in healthcare depends on the responsible management of sensitive patient data. Safeguarding confidentiality and maintaining patient trust require protection against unauthorised access, misuse and data breaches (Tariq and Hackert, 2023; Conduah *et al.,* 2025). Technological measures such as encryption and privacy-preserving techniques can enhance data security (Torkzadehmahani *et al.,* 2022).

While AI can strengthen data security, its reliance on large volumes of sensitive data heightens the risk of breaches and ethical violations, making strong governance essential. To mitigate these risks, regulatory frameworks such as the Health Insurance Portability and Accountability Act of 1996 (HIPAA), provide guidance on the lawful handling of patient information, permitting limited disclosures without consent while protecting the integrity of medical records (Tariq and Hackert, 2023).

Discussion

This project successfully compared the performance of a custom CNN, with a CNN encompassing a transfer learning approach based on ResNet50V2, for multi-class AD classification using structural MRI images. The transfer learning approach yielded better performance in terms of classification accuracy and reliability, highlighting the advantage of pre-trained architectures in limited medical imaging datasets

The results align with prior research, demonstrating the effectiveness of transfer learning in Alzheimer's MRI classification. Krishnapriya and Karuna (2023), evaluated pre-trained models such as ResNet50, VGG-19 and Inception V3 to categorise MRI images. They found that VGG-19, with transfer learning, achieved highest performance (99.48% accuracy), outperforming other models while requiring fewer epochs and limited data. These findings support the idea that pre-trained CNNs can efficiently generalise to medical imaging tasks. Similarly, the improved performance observed in Model 2 reinforces the value of transfer learning in AD classification.

Another strength of this study was the use of ReLU activation function in both models to mitigate the vanishing gradient problem. Vanishing gradients occur when feedback signals diminish as they propagate backward through deep networks, causing early layers to learn slowly or stop learning entirely (Yilmaz and Poli, 2022). Unlike sigmoid functions, ReLU maintains stronger gradients which enable more efficient training of deep image classification models. This likely contributed to more efficient optimisation during training (Ntambwe, 2025). Additionally, CLAHE preprocessing enhanced contrast and improved feature separability of the MRI images, potentially contributing to improved model performance.

Gradients play a central role in neural network optimisation, dictating how weights are adjusted during backpropagation (Brownlee, 2021). However, unstable training behaviour can occur when gradients become excessively large, leading to erratic weight updates and fluctuation in accuracy/loss curves. This phenomenon is commonly referred to as "exploding gradient" (Brownlee, 2019; GeeksforGeeks, 2026). The instability seen in Model 1 may reflect gradient this instability.

Similar observations were reported by Zhang *et al.,* (2024), which introduced an improved VGG16-based AD classification model (ADNet) using data from the "Alzheimer's Disease Neuroimaging Initiative" (ADNI). The researchers discovered that using the ReLU activation function caused large, unstable fluctuations in their loss curves, which they identified as gradient explosions destabilising their model. The authors discovered that replacing ReLU with the Exponential Linear Unit (ELU) significantly improved training stability and overall diagnostic performance. This implies that activation function choice can influence optimisation stability in deep learning models.

Despite Model 2's performance, evidence of mild overfitting was present. Training accuracy continued to improve while validation performance plateaued, suggesting the model may have begun memorising patterns instead of learning. This can be likely due to the small dataset, as the model's high capacity of trainable parameters could've led to the memorisation of training features and noise instead (Zhang *et al.,* 2022).

Another limitation concerns the use of two-dimensional JPG MRI slices. AD involves three-dimensional brain structures, and 2D images may limit the model's ability to holistically capture atrophy in key regions such as hippocampus (Breijyeh and Karaman, 2020; Liu *et al.,* 2022)

Future work should incorporate tools like Gradient-weighted Class Activation Mapping (Grad-CAM), to highlight brain regions that're most strongly influencing model predictions (Zhang *et al.,* 2021). Implementing ELU in replacement of ReLU activation may also improve training stability and overall model robustness in classification.

Conclusion

This project developed and evaluated two CNN architectures for classifying AD severity, using 5,076 MRI scans. The findings suggest that the transfer learning approach (ResNet50V2) significantly outperformed the custom CNN, achieving higher overall F1-scores across all disease states, and a superior discriminative performance (AUC=0.933). These results reinforce the value of transfer learning in medical imaging analysis.

Despite its strong performance, evidence of overfitting (Model 2) and training instability (Model 1) highlights the need for further optimisation of these models before being implemented to a healthcare setting. Future work should entail the use of larger and more diverse datasets, alternative activation functions, gradient stabilisation techniques and explainability tools like Grad-CAM to enhance transparency.

In conclusion, while not yet suitable for independent clinical deployment, this project highlights the potential of AI-assisted MRI analysis to augment diagnostic workflows in Alzheimer's disease.

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